

Ryan Campbell

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Research Interests

Multivariate and spatio-temporal extreme value theory, dependence modelling, applications to real world data.

Professional Activities

- 2025–2027 Postdoctoral researcher, Université Côte d'Azur
Nice, France
- 2020–2021 Data science intern, Desjardins General Insurance Group
Lévis, Québec

Education

- 2021–2025 PhD Statistics, Lancaster University
Thesis: A geometric interpretation of multivariate extreme value analysis
Supervisor: Jennifer L. Wadsworth
- 2019–2020 MSc Mathematics & Statistics, McGill University
Thesis: Deterministic Gaussian averaged neural networks
Supervisor: Adam Oberman
- 2015–2018 BSc Mathematics, McGill University
Project: Semiparametric modelling of max-stable processes using Kendall's tau
Supervisor: Johanna Nešlehová

Publications

- [1] **R. Campbell**, K Grolmusova, L. Kakampakou, and J. Lee. Analysing Extreme Rainfall via a Geometric Framework, *Extremes*. To appear.
- [2] L.M. André, **R. Campbell**, E. D'Arcy, A. Farrell, D. Healy, L. Kakampakou, C. Murphy, C.J.R. Murphy-Barltrop, M. Speers. Extreme value methods for estimating rare events in Utopia. *Extremes*, 1–23, 2024.
- [3] J. L. Wadsworth and **R. Campbell**. Statistical inference for multivariate extremes via a geometric approach. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, (86)5: 1243–1265, 2024.

Preprints

- [1] **R. Campbell** and J. L. Wadsworth. Piecewise-linear modeling of multivariate geometric extremes. *arXiv:2412.05195*, 2024.
- [2] I. Papastathopoulos, L. de Monte, **R. Campbell**, H. Rue. Statistical inference for radially-stable generalized Pareto distributions and return level-sets in geometric extremes. *arXiv:2310.06130*, 2023.
- [3] **R. Campbell**, C. Finlay, and A. M. Oberman. Adversarial Boot Camp: label free certified robustness in one epoch. *arXiv:2010.02508*, 2020.
- [4] **R. Campbell**, C. Finlay, and A. M. Oberman. Deterministic Gaussian averaged neural networks. *arXiv:2006.06061*, 2020.

Presentations

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| June 2025 | 14 th International Conference on Extreme Value Analysis
Title: Geometric spatio-temporal extremes
<i>with Kristina Grolmusova, Lydia Kakampakou, and Jeongjin Lee.</i>
Location: University of North Carolina, Chapel Hill, NC, USA |
| June 2025 | 14 th International Conference on Extreme Value Analysis
Title: Piecewise-linear modeling of multivariate geometric extremes
Location: University of North Carolina, Chapel Hill, NC, USA |
| Nov. 2024 | Various statistics seminars in France and Switzerland
Title: A geometric approach to multivariate extremal inference
Locations: Inria Grenoble, Université de Genève, École Polytechnique Fédérale de Lausanne, Université Claude Bernard Lyon 1, INRAE Avignon. |
| Sept. 2024 | UQÀM Statistics Seminar
Title: New developments for a geometric approach to multivariate extremal inference
Location: UQÀM, Montréal, QC, Canada |
| Oct. 2023 | HEC Statistics Seminar
Title: Statistical inference for multivariate extremes via a geometric approach
Location: HEC Montréal, Montréal, QC, Canada |
| Sept. 2023 | STOR-i Extremes Workshop (STEW)
Title: Modelling extremal dependence of a 3-dimensional oceanographic dataset via a semi-parametric geometric approach
Location: Lancaster University, Lancaster, UK |
| June 2023 | 13 th International Conference on Extreme Value Analysis
Title: A geometric approach for modelling negative asymptotic dependence
Location: Bocconi University, Milan, Italy |

Teaching Assistant Positions

Lancaster University

Winter 2024	MATH333	Statistical Models (GLMs)
Winter 2024	MATH113	Convergence and Continuity
Winter 2023	MATH140	Statistics
Winter 2022	MATH456/556	Extreme Value Theory
Winter 2022	MATH235	Statistics II
Fall 2021	MATH330	Likelihood Inference

McGill University

Fall 2020	MATH208	Intro. to Statistical Computing
Fall 2019	MATH597	Topics in Applied Mathematics: Mathematics of Machine Learning
Fall 2019	MATH223	Linear Algebra

Awards and Funding

2023–2025	NSERC Postgraduate Scholarship-Doctoral	CA\$63,000
2023–2025	FRQNT Doctoral Research Scholarship	CA\$25,334
2021–2025	EPSRC Mathematical Sciences studentship	£62,436 (minimum)
2025	London Mathematical Society Early Career Researcher Travel Grant	£500
2025	Institute of Mathematics Small Grant	£600
2025	Graduate College & Doctoral Academy Travel Grant (Lancaster University)	£200
2023	Nick Smith Prize (Lancaster University)	£500
2020	Mitacs internship at Desjardins	CA\$13,000
2019–2020	Master's degree funding	CA\$20,500
2019–2020	McGill University Graduate Excellence Award	CA\$3,400
2018	Science Undergraduate Research Award	CA\$6,500

Languages and Skills

- Fluent in English and French.
- Proficient in R, Python (incl. PyTorch), Matlab, LaTeX, Java, HTML, Linux

Extracurricular Activities

- **Treasurer**, Bowland Ultra
Lancaster, UK
2023–2025
- **Treasurer**, Lancaster University Folk Society
Lancaster University
2023–2025 academic years.

- **VP Finance**, Graduate Student Association for Mathematics and Statistics (GSAMS)
McGill University
2019–2020 academic year.
- **Volunteer at the 2018 Statistical Society of Canada annual meeting**
Roles: Setting up audio-visual equipment for presentations and directing conference attendees to presentations.
Location: McGill University
3–6 June 2018